

ALEXANDER MA

✉ alexanderma806@gmail.com ☎ 806-881-9570 🌐 alexander-ma 📺 ma-alex

EDUCATION

The University of Texas at Austin

May 2019

B.S. Electrical and Computer Engineering, GPA: 3.19/4.00

Coursework – *Algorithms, Data Structures, Software Design, Data Science, Concurrent & Distributed Systems*

EXPERIENCE

Ovation.io – Technical Staff, Software Engineering Intern; Cambridge, MA

June 2018 – August 2018

- Worked on a Lab Information Management System to track and manage lab data using Ember.js and Ruby on Rails
- Refactored backend asynchronous background tasks to use in-memory data store and multi-threaded processing
- Improved scalability and performance within the platform and reduced MySQL database garbage collection

Longhorn Center for Academic Excellence – Web Administrator Intern; Austin, TX

September 2017 – May 2018

- Managed web pages using WordPress to help students obtain access to tutoring and on-campus resources
- Collaborated with staff members to improve user experience and updated content based on staff needs
- Integrated extensions and plugins that help track user analytics in order to see what students view

SKILLS

Programming Languages: Java, C, Python, Android, JavaScript, HTML, CSS, Ruby, Assembly

Frameworks & Tools: Git, jQuery, Bash, Z Shell, Ember.js, Ruby on Rails, Jira, Hadoop

Operating Systems: OS X, Windows, Linux

Spoken Languages: English, Cantonese, Spanish

PROJECTS

Drake Lyrics Generator – Data Science Lab Project

April 2018

- Produced Drake-like rap lyrics using N-Grams and Neural Nets in order to create an artificial ghostwriter
- Scraped artist lyrics from Genius.com and performed data analysis using BeautifulSoup and Python

Transient – Web Messaging System Project

December 2017

- Created an application for users to anonymously create, share, and destroy transient chat room channels
- Developed using jQuery and Google Firebase to manipulate channels, store messages and user information

Safely – HackTX: JP Morgan Challenge Winner: Best Use of Social Media Data for Disaster Response

October 2017

- Developed an Android app to assist people in visualizing those impacted by natural disasters and its severity
- Implemented Google Maps API and Google Firebase to store incident information and display incidents worldwide

Chat Client – Java Programming Project

April 2017

- Developed a chat client using Java, JavaFX, reflection, SQL database, and socket programming
- Implemented features such as private/group messages from different machines, emoticons, and user login

Pole Position – Embedded Systems Video Game Project, Finalist

May 2016

- Programmed an obstacle racing game onto a T.I. microcontroller with a partner using C and Assembly
- Constructed graphics to display on an Adafruit ST7735 LCD and implemented a 4-bit DAC for sound effects

LEADERSHIP & ACTIVITIES

Freetail Hackers – Organizer

January 2016 – December 2017

- Organized annual hackathons that challenged students in computer programming to create innovative projects

CodePath University – Web Security Student

January 2017 – May 2017

- Learned cybersecurity fundamentals involving concepts such as penetration testing and encryption